IE3035 Operations Research

HW3

Q1. Consider the following LP:

Maximize $z = 3x_1 + 2x_2$

subject to

$4x_1 - x_2 \leq 8$

$4x_1 + 3x_2 \leq 12$

$4x_1 + x_2 \leq 8$

$x_1, x_2 \geq 0$

(a) Show that the associated simplex iterations are temporarily degenerate.

(b) Verify the result by solving the problem graphically.

Q2. Solve the following LP:

Maximize $z = 2x_1 - x_2 + 3x_3$

subject to

$x_1 - x_2 + 5x_3 \leq 10$

$2x_1 - x_2 + 3x_3 \leq 40$

$x_1, x_2, x_3 \geq 0$

From the optimal tableau, show that all the alternative optima are not corner points (i.e., nonbasic).

Q3. Consider the following LP:

Maximize $z = 2x_1 + x_2$

subject to

$x_1 - x_2 \leq 10$

$2x_1 - x_2 \leq 40$

$x_1, x_2 \geq 0$

Solve the problem by simplex algorithm. Is there a special case for this problem? If there is, what is it? Explain.